Vina docking-score matrices are strongly spectrally concentrated before, but not after, two-way centering

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Abstract

Docking scores are commonly interpreted one target at a time, although reverse docking, off-target profiling and multiobjective molecular design use them as a ligand-by-target matrix. We quantified variance concentration in two large AutoDock Vina matrices using the participation-ratio (PR) effective dimension of the column-standardized target-correlation spectrum. A 12,651-compound by 44-target panel had PR dimension 1.83 (principal component 1, 73%), and DOCKSTRING, with 260,060 complete molecules by 58 targets, had PR dimension 2.27 (principal component 1, 66%). Leave-one-target-out values were 1.76–1.85 and 2.22–2.31, respectively; random and target-family-stratified subsampling retained the concentration. A Bemis–Murcko scaffold-cluster bootstrap on a fixed DOCKSTRING chemical support gave a 95% interval of 2.15–2.37. Removing ligand and target main effects before applying the same column standardization changed the estimand and increased PR dimension to 9.30 and 18.21. Five hundred column-permutation nulls yielded 3 versus 7 and 4 versus 10 supra-null modes before versus after two-way centering, with identical counts in five independent 100-permutation series. Two small RF-Score v1 blocks supported the direction but were too small to generalize beyond the tested Vina systems. On a matched ChEMBL surface of 74 compounds by six targets, exact-relation median aggregation gave PR dimensions 3.83 for experiment and 1.83 for docking; human binding and Ki/Kd restrictions preserved the ordering. This comparison supports the docking result on that matched support but does not characterize experimental affinity matrices in general. Strong spectral concentration of the column-standardized Vina surfaces therefore coexists with a substantially richer two-way-centered residual structure.

Scientific Contribution. We provide a preprocessing-matched spectral analysis of two large

dense Vina panels, including uncertainty with respect to chemical scaffolds and target-panel composition. By separating the column-standardized score surface from its two-way-centered residual, we show why low aggregate PR dimension cannot be interpreted as absence of target-specific variation.

Keywords: molecular docking; AutoDock Vina; effective dimension; spectral analysis; two-way centering; reverse docking; target profiling.

1 Introduction

A docking score is scalar for one protein–ligand pose, but many applications produce a matrix. Reverse docking ranks targets for a ligand, secondary-pharmacology panels compare off-target liabilities, and multiobjective design uses scores against several desired and undesired proteins. In each case the useful object is a matrix $X \in \mathbb{R}^{N \times P}$ whose rows are ligands and columns are targets. Interpreting its P columns as P independent target-resolved descriptors assumes that the matrix actually contains a commensurate number of independent directions.

Two long-recognized biases make this assumption doubtful. Empirical energy functions often reward larger ligands because additional atoms contribute additional favorable contacts; normalization by molecular weight, heavy-atom count or learned physicochemical corrections can change virtual-screening rankings.[4, 5, 6] Across proteins, scores also depend on pocket contact area, size and hydrophobicity. In a reverse-docking study of 5,000 molecules and 100 proteins, these target properties produced recurrent “interference proteins” and motivated per-target normalization.[7] These observations identify ligand and target main effects. They do not say how much of a complete score matrix those effects occupy, how many collective directions remain, or whether a low-dimensional column-standardized surface is equivalent to poor selectivity.

Two-directional normalization and principal-component analysis of docking matrices also precede this study. SePreSA applied a ligand-wise Z transformation followed by normalization of protein-score vectors to a 79-compound by 86-protein docking matrix. [1] Fukunishi and colleagues applied PCA to a complete docking matrix for compound and pocket classification, and later used multi-protein docking-score profiles as QSAR descriptors.[2, 3] These studies establish that matrix-wide normalization and PCA are useful. The contribution here is narrower: a large-scale quantitative estimate of spectral concentration, explicit comparison of column-standardized and two-way-centered estimands, and uncertainty analyses for chemical and target-panel composition.

The distinction matters because docking has several separable tasks. A method may reproduce a crystallographic pose yet rank affinities poorly, and no single program performs uniformly across targets and evaluation objectives.[8] Likewise, the raw variance of a score matrix and the correctness of within-ligand target contrasts are different objects. Writing

$$X_{ij} = \mu + a_i + b_j + \gamma_{ij}, \tag{1}$$

separates the grand mean μ, the ligand tendency a_i to score well across targets, the target tendency b_j to score ligands favorably, and the interaction γ_{ij}. The column-standardized spectrum is allowed to be dominated by a_i. Selectivity, however, depends on differences within rows and ultimately on γ_{ij}; the additive terms cancel from a double difference. A rank estimate on X and the same estimate on γ therefore answer different questions. Established panel selectivity summaries such as Gini, entropy and window scores also answer different questions and can reorder the same compounds, making the metric definition part of the estimand rather than a cosmetic choice.[23, 24]

Complete public score matrices now permit this distinction to be tested at useful scale. DOCKSTRING contains docking scores for more than 260,000 molecules against 58 medically relevant targets.[9] An independent acute-toxicity resource contains 12,651 analyzable compounds docked against a 44-protein secondary-pharmacology panel.[10] We combine these matrices with two small RF-Score rescoring panels and matched experimental affinity data. RF-Score replaces a predetermined empirical functional form with a random-forest model.[11] The RF-Score blocks are supporting sensitivities rather than independent large-scale replications.

Low-rank representations are not new to chemogenomics. Kernelized Bayesian matrix factorization, probabilistic matrix factorization and neighborhood-regularized latent-factor models deliberately embed sparse DTI observations in lower-dimensional spaces to improve prediction or matrix completion.[17, 18, 19] Our question is different: we do not impose a low-rank predictive model on sparse experimental data. We measure the empirical spectrum of dense docking output, then ask which part is attributable to additive ligand and target effects. The two-way decomposition connects this diagnostic to classical additive-model reasoning, while participation and entropy ranks provide continuous summaries rather than a chosen latent dimension.[20, 21]

We ask three questions. First, do the two large tested Vina panels concentrate their variance into roughly two PR effective dimensions under a common estimator and empirical null? Second, is the observation stable to changes in chemical support and target-panel composition, and is it directionally supported by the deliberately limited RF-Score blocks? Third, what remains after ligand and target main effects are removed? The essential boundary is that two-way-centered residual surfaces can be much richer than their column-standardized matrices; aggregate concentration therefore does not establish absence of target-specific variation.

2 Results

2.1 Two independent Vina panels have concentrated column-standardized spectra

The 12,651-by-44 antitarget panel was analyzed after clipping the rare positive Vina scores to zero, mean-imputing missing entries and standardizing each target column. The largest eigenvalue accounted for 73.3% of the correlation spectrum. The PR effective dimension,

$$r_{\mathrm{PR}} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2}, \tag{2}$$

has a direct correlation interpretation. Because a P-target correlation matrix has $\mathrm{tr}(C) = P$,

$$r_{\mathrm{PR}} = \frac{P^2}{\mathrm{tr}(C^2)} = \frac{P}{1 + (P-1)\overline{r_{ij}^2}}, \tag{3}$$

where the bar is over unique off-diagonal target pairs. Thus, a small PR dimension is an alternative summary of strong mean-squared inter-target correlation; the full spectrum additionally shows how that dependence is distributed among collective modes. The estimated PR dimension was 1.834, with a Butina-cluster bootstrap interval of 1.785–1.889 (Fig. 1). Three leading modes exceeded the rank-wise 95% column-permutation null under the common 12,000-row, 500-permutation parallel-analysis protocol. The matrix is not algebraically rank two: eight principal components are needed to reach 90% of variance, and the entropy effective rank is 3.90. "Collapse toward two" therefore refers specifically to participation-ratio variance concentration. Full column-standardized and residual scree plots are given in Supplementary Fig. S1.

The 111 zero-censored cells did not drive this concentration. Treating them as missing and target-mean imputing gave a PR dimension of 1.829, while excluding every affected ligand gave 1.814 (12,544 ligands). Excluding the four most-censored targets gave 1.602, but this changes the target estimand and is interpreted as panel-composition sensitivity rather than an imputation effect (Supplementary Table S2). A chemical-cluster block-permutation null and a Marchenko–Pastur edge based on the 8,150 effective clusters both retained three leading modes, matching the independent-column null.

The independent DOCKSTRING matrix gave the same qualitative result at much larger N and with 58 targets. Among 260,060 molecules with complete scores, principal component 1 accounted for 65.9% of the spectrum and r_{PR} was 2.266 (molecule-bootstrap interval 2.239–2.298). On a fixed, seeded support of 15,000 molecules, the ordinary molecule-bootstrap interval was 2.230–2.296 and the Bemis–Murcko scaffold-cluster interval was 2.148–2.367. Four leading modes exceeded the same common-protocol null. Its entropy rank was 6.21 and 22 components were needed for 90% of variance, again distinguishing a concentrated spectrum from literal rank deficiency. The narrow interval is conditional on the fixed 58-target panel and on molecule-level resampling; it does not quantify target-panel uncertainty or extrapolation to another chemical population.

The result was also stable to the observed target-panel composition. Leave-one-target-out PR dimensions ranged from 1.756 to 1.853 for Docking-44 and from 2.218 to 2.311 for DOCKSTRING-58 (Supplementary Fig. S2). Random target subsets showed saturation rather than a linear increase: at 20 targets the median column-standardized PR dimensions were 1.79 and 2.18, compared with 1.83 and 2.27 for the full panels (Fig. 1c). Broad target-family-stratified sampling gave similar 20-target medians of 1.85 and 2.22. For the two-way-centered residuals, the corresponding family-stratified medians were 7.38 and 11.40, increasing to 9.30 and 18.21 in the full panels (Fig. 1d). These subset distributions quantify sensitivity within the tested biological panels; the target lists are not probability samples from a universal target population.

2.2 Small RF-Score panels and target expansion provide preliminary sensitivities

In two small blocks, a scoring-function change did not restore a high-dimensional column-standardized surface. RF-Score v1 applied to poses across 14 targets gave $r_{\mathrm{PR}} = 1.416$ on the 41-ligand common block (bootstrap interval 1.25–1.70; principal component 1, 83.7%). Vina scores on the same support had PR dimension 1.25. Because those poses were sampled with smina/Vina, this comparison isolates the scoring function but not pose generation.

To separate the result from Vina pose generation, we repeated RF-Score on poses generated with the Vinardo function across 20 targets. The 35-ligand complete block remained concentrated: $r_{\mathrm{PR}} = 1.812$ (1.472–2.439) and principal component 1 accounted for 73.7%. The corresponding

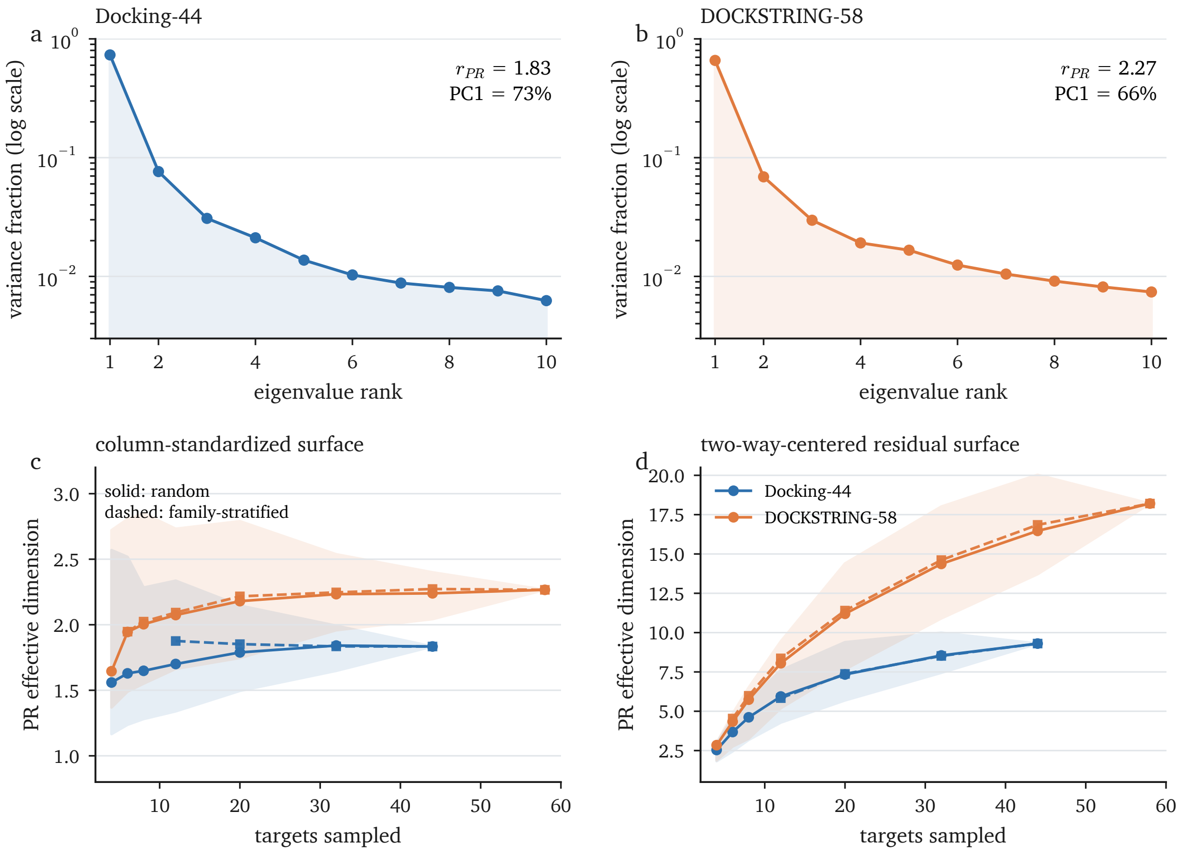

Figure 1: Two independent Vina panels have concentrated column-standardized spectra. a,b, Eigenvalue spectra and PR effective dimensions of the 44-target antitarget and 58-target DOCKSTRING matrices after column standardization. **c,d**, Column-standardized and two-way-centered residual PR effective dimensions under target subsampling. Solid lines and shaded 2.5–97.5% intervals show 200 uniform subsets; dashed square-marked lines show 200 family-stratified subsets where the requested size permits representation of every broad family. Full-panel points use all ligands.

original Vina scores had PR dimension 1.29, and the two leading-score directions aligned at $|r| = 0.81$. The result is a preliminary sensitivity covering one older learned scoring function with wide uncertainty, not a general replication across modern CNN- or GNN-based scorers.

Nor did adding purpose-selected targets to the antitarget panel create an independent column-standardized dimension. Adding BSEP, FXR and PXR increased the target count from 44 to 47, but changed r_{PR} only from 1.834 to 1.877 and retained three supra-null modes. Their correlations with the pre-existing per-ligand panel mean were 0.92, 0.53 and 0.81, respectively. Under this representation, new columns can largely project onto an existing shared docking-score direction.

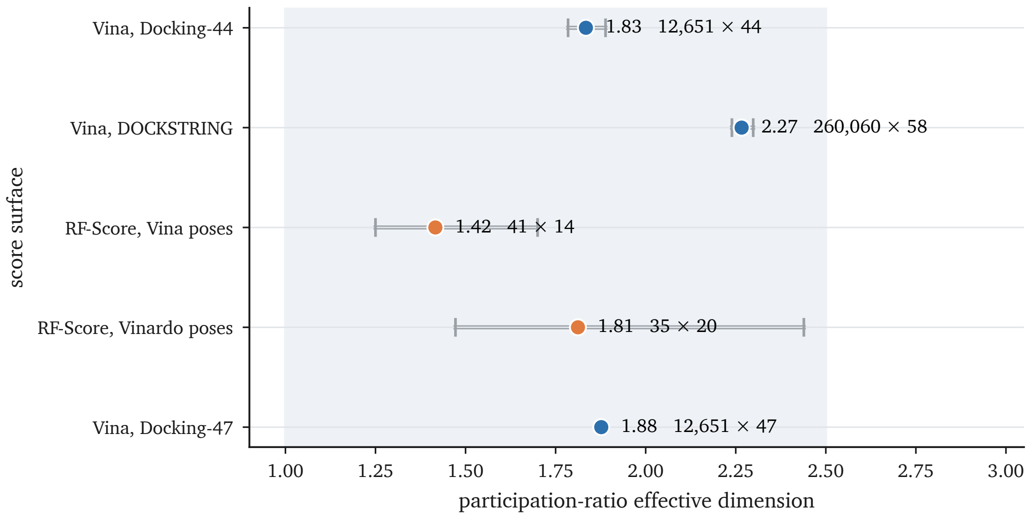

Figure 2: Low column-standardized PR dimension persists across tested settings. PR effective dimension and bootstrap intervals are shown for the two large Vina panels, RF-Score on Vina poses, RF-Score on Vinardo poses and the 44-to-47-target panel expansion. Point labels report ligand and target counts. Intervals use the bootstrap protocol of the source analysis; the 47-target expansion is a deterministic sensitivity and is shown without an interval.

Table 1: Summary of the score and affinity matrices. Missingness is reported on the final analysis block immediately before primary imputation; complete-common blocks have zero missingness by construction. Uncertainty procedures are conditional on the listed target panel. Full preprocessing and provenance are given in Supplementary Tables S1–S7.

Matrix	Ligands	Targets	Missing	Primary preprocessing	Column-standardized PR / entropy	Residual PR
Docking-44	12,651	44	1.466%	target mean; column z-score	1.83 / 3.90	9.30
DOCKSTRING-58	260,060	58	0.0022%	complete rows; column z-score	2.27 / 6.21	18.21
RF-Score, Vina poses	41	14	0	complete common block	1.42 / 2.16	6.68
RF-Score, Vinardo poses	35	20	0	complete common block	1.81 / 3.35	9.92
Matched docking	74	6	0	complete matched block	1.83 / 2.60	3.65
Matched ChEMBL	74	6	9.91%	exact relation; median aggregation	3.83 / 4.63	4.13

2.3 The leading direction is a shared docking-score axis

The dominant direction was shared across targets rather than localized to one receptor family. In DOCKSTRING, the absolute correlation between principal-component 1 scores and the per-ligand mean docking score was 0.995; the median absolute inter-target correlation was 0.706. In the antitarget panel, seven physicochemical descriptors explained 60.6% of the variance in the per-ligand mean docking score. Heavy-atom count alone correlated -0.43 with that mean, consistent with favorable additive contributions as ligand size grows.

Removing size-related information weakened but did not erase the concentration. Linear residualization of each target score against seven physicochemical descriptors increased r_{PR} from 1.83 to 2.54; cross-fitted nonlinear residualization increased it to 3.52. Restricting the data to 4,308 molecules with 17–22 heavy atoms still gave PR dimension 1.70. Dividing scores by heavy-atom count made the spectrum more concentrated (PR dimension 1.14), showing that ligand-efficiency normalization is not a dimensionality correction. Across 36 targets, pocket volume correlated 0.514 (target bootstrap interval 0.272–0.686) with the absolute correlation between each target and the per-ligand panel mean. A 13-descriptor pocket model had negative leave-one-target-out R^2 for both ordinary least squares (-0.011) and ridge regression (-0.144), so the data do not support a multivariable pocket-mechanism claim. On the exact 74-by-6 experimental support, linear and five-fold cross-fitted nonlinear descriptor residualization raised docking PR dimension only to 2.81 and 2.68, respectively, still below the exact-relation median experimental value of 3.83. Thus the matched contrast is not solely a consequence of the seven measured bulk descriptors.

2.4 A matched ChEMBL surface is less concentrated than docking on the same support

A concentrated docking surface could reflect its compounds or targets rather than its score model. We therefore used the same 74 compounds and six aminergic targets for docking and ChEMBL. The primary experimental curation required an exact standard relation, retained Ki, Kd, IC50 and EC50, aggregated repeated compound–target activities by the median, and required measurements for at least five of six targets. Target-median imputation filled 9.9% of cells. The resulting ChEMBL surface had PR dimension 3.833, compared with 1.827 for docking on the same support (Fig. 4).

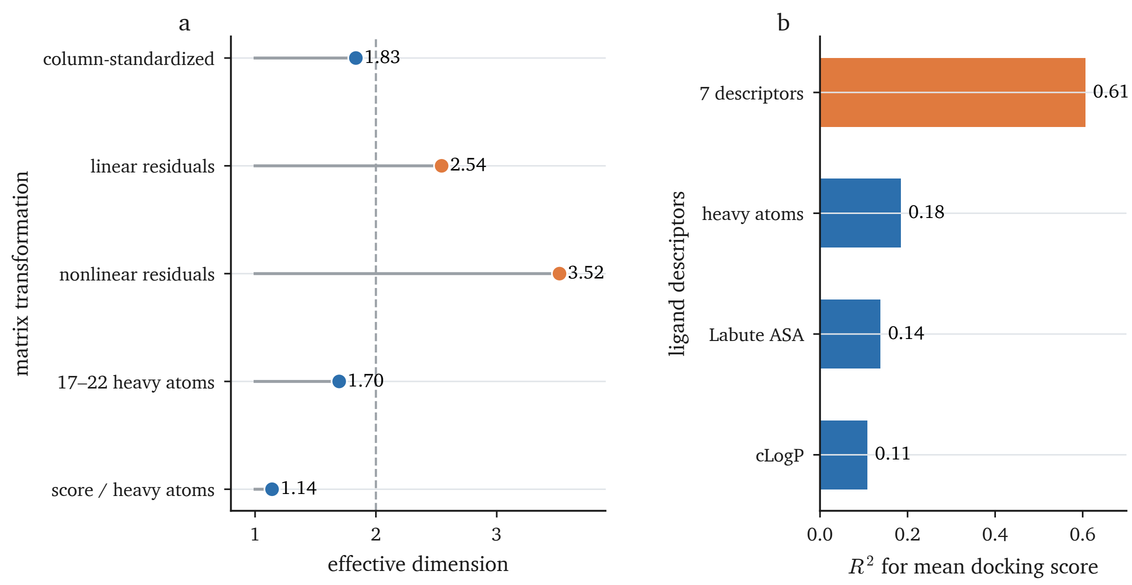

Figure 3: Bulk ligand properties explain part of the shared docking-score axis. **a**, PR effective dimension for the column-standardized antitarget matrix, linear and cross-fitted nonlinear physicochemical residuals, a narrow heavy-atom band and ligand-efficiency normalization. **b**, Variance in the per-ligand mean docking score explained by the seven-descriptor model and single-descriptor controls. The residualized matrices are diagnostic transformations, not proposed scoring functions.

More restrictive assay blocks preserved the ordering. Human binding assays with all four endpoints gave 3.510 for ChEMBL and 1.865 for docking on 47 ligands. Restricting further to human binding Ki/Kd gave 3.072 and 1.894 on 35 ligands. These comparisons do not define the dimension of experimental binding in general: the assays come from multiple laboratories and conditions, up to 13.3% of cells are imputed, and six targets cap the centered algebraic dimension at five. They support only the statement that these matched ChEMBL surfaces are less spectrally concentrated than docking on the same ligand–target support.

Additional controls addressed missingness and measurement heterogeneity. A fully observed 31-by-6 ChEMBL sub-block had PR dimension 3.61. On a secondary 212-by-22 block, estimates were 5.72 with K-nearest-neighbour imputation, 8.59 averaged over chained imputations, 2.64 with low-rank matrix completion and 7.28 with probabilistic-PCA expectation maximization; the magnitude depended on the estimator. A separate frozen ChEMBL extraction with the same nominal 212-by-22 shape and 52.9% missingness gave KNN PR dimension 6.85; we retain both in Supplementary Table S3 as a curation-sensitivity discrepancy. A legacy most-potent aggregation of the 74-by-6 block gave 3.734 for experiment and 1.827 for docking, with a paired bootstrap difference interval of 1.05–2.61, and is retained only as a sensitivity. Different-document repeats estimated a standard deviation of 0.493 pChEMBL units. Adding target-scaled Gaussian noise at that level to docking raised its median PR dimension to 2.29 (simulation interval 2.12–2.50); approximately 2.5 times that scale was required to reach the legacy experimental value. None of these controls removes all biological, laboratory or condition-related heterogeneity.

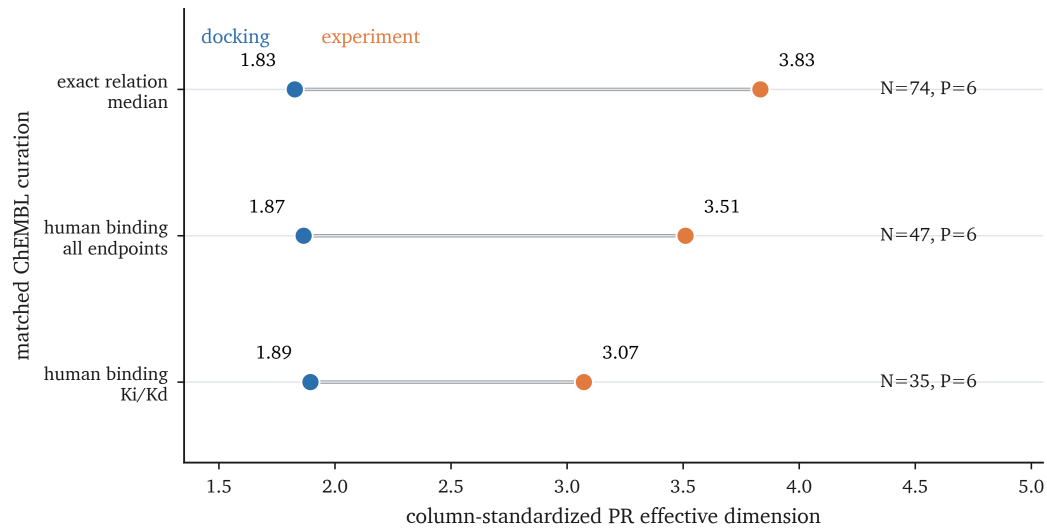

Figure 4: Matched ChEMBL surfaces are less concentrated than docking. PR effective dimensions are shown for exact-relation median aggregation on the 74-by-6 support, human binding assays with all endpoints on 47-by-6 support, and human binding Ki/Kd on 35-by-6 support. Each line connects docking and experiment on identical compounds and targets; labels report ligand and target counts.

2.5 Two-way centering reveals higher-dimensional residual surfaces

The column-standardized spectrum mixes the ligand main effect with the ligand–target residual. We removed both row and column means according to Eq. 1 and then applied the same column standardization and spectral estimator. This changed the measured object and the answer (Fig. 5a). The 44-target Vina panel rose from PR dimension 1.834 to 9.303; principal component 1 fell from 73.3% to 27.4%. DOCKSTRING rose from 2.266 to 18.206; principal component 1 fell from 65.9% to 17.8%. The two RF-Score panels rose from 1.416 to 6.681 and from 1.812 to 9.918. In unstandardized sum-of-squares terms, the interaction residual retained 27% of grand-centered variation in the antitarget panel and 31% in DOCKSTRING. Low-variance relative to a main effect is not the same as absent.

The increase was not only a rescaling artifact of the PR denominator. Relative to the available algebraic maximum, normalized PR dimensions increased from 0.042 ($1.834/44$) to 0.216 ($9.303/43$) for Docking-44 and from 0.039 ($2.266/58$) to 0.319 ($18.206/57$) for DOCKSTRING. This normalization is a ceiling reference, not a correction for target count: the residual PR dimension increased with panel size while its ratio to $P-1$ decreased (Fig. 1d). At a matched 44 targets, the DOCKSTRING median residual PR dimension was 16.47 (target-subset interval 13.72–20.03), compared with 9.30 for the full Docking-44 panel. The cross-dataset difference therefore remained at matched P, although part of the 18.21-versus-9.30 contrast reflects different target counts.

Under an additive surface with independent identically distributed residual noise, two-way centering produces a residual correlation spectrum with PR dimension approaching the $P-1$ ceiling. The observed values lie well below that ceiling, so the residuals remain correlated rather than resembling independent noise. Under a common 12,000-row, 500-permutation protocol, modes above the rank-wise 95% null increased from 3 to 7 and from 4 to 10, respectively. The counts were identical in five independent 100-permutation series for each matrix and transformation. The null applies the complete column-standardized or centered transformation after independently permuting every column, so it tests each estimand in its own constrained space.

Two-way centering also creates an exact structural constraint: every interaction row sums to zero. Non-zero column scaling preserves one linear dependence, so the interaction matrix has algebraic rank at most $P-1$ and one correlation eigenvalue is zero. Residual and column-standardized PR dimensions therefore describe different constrained spaces; the centered increases occur despite, not because of, a larger algebraic maximum.

The same distinction appeared in learned-affinity surfaces. On the common DAVIS evaluation surface, experimental affinity had column-standardized and residual PR dimensions of 7.448 and 13.761. Boltz-2 had a lower column-standardized value of 3.259 but a residual value of 15.485. The latter slightly exceeded the experimental value, yet Boltz-2 did not reproduce experimental selectivity perfectly. Residual PR dimension measures how many directions vary, not whether their geometry matches experiment.

3 Discussion

The two tested large Vina panels give a consistent result: dozens of target columns occupy about two PR effective dimensions after column standardization. This statement remained stable under leave-one-target-out deletion, uniform and family-stratified target subsampling, and chemical resampling. It is nevertheless estimator-specific. Entropy effective ranks and the numbers of

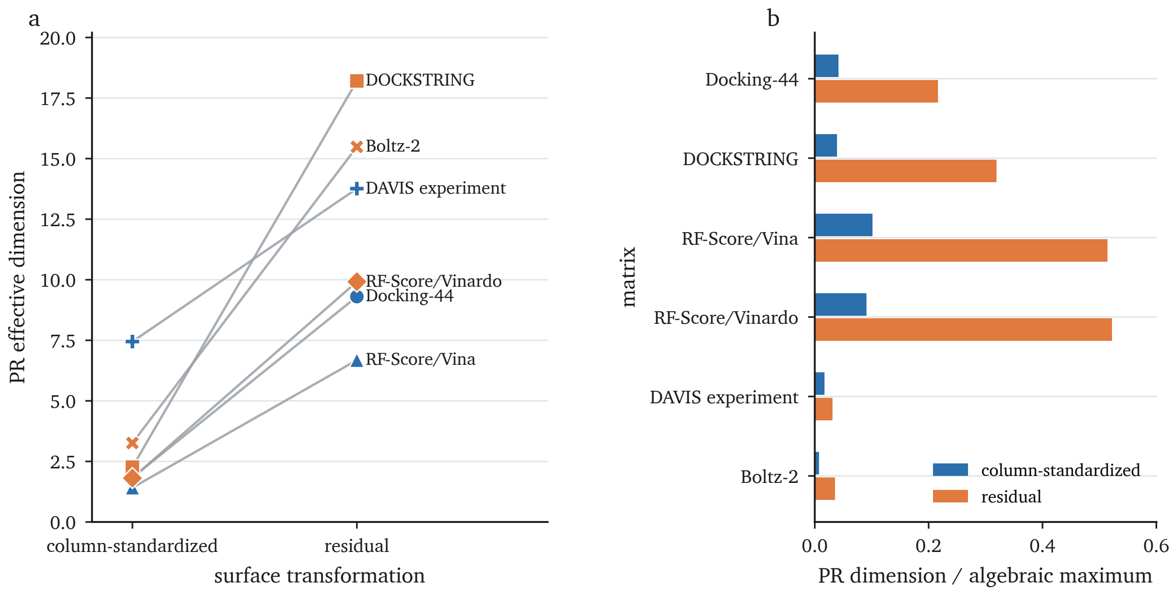

Figure 5: Main-effect removal uncovers additional residual dimensions. **a**, Column-standardized and two-way-centered residual PR effective dimensions for Vina, DOCKSTRING, two small RF-Score blocks, experimental DAVIS affinity and Boltz-2. Lines connect the same matrix under the two transformations. **b**, Each PR value divided by its algebraic maximum, P before and $P-1$ after centering. The DAVIS and Boltz-2 surfaces provide contextual learned-affinity examples rather than independent docking replications.

components required for 90% variance were larger, so "dimension two" must not be detached from the PR definition. The observed phenomenon is strong spectral concentration, equivalently high mean-squared inter-target correlation under Eq. 3, rather than algebraic rank deficiency.

The novelty is not two-way normalization or PCA itself. SePreSA used two-directional normalization of docking-score matrices, and earlier protein–compound affinity work used principal components as multidimensional descriptors.[1, 2, 3] Here, the added information is a quantitative comparison of spectral concentration on two large dense panels, the separation of column-standardized and two-way-centered estimands, and explicit sensitivity to target-panel and chemical composition.

The shared docking-score axis provides an explanation at the level supported by these data. Ligand bulk affects many additive score terms in the same direction, and pocket properties shift targets systematically. Prior normalization work identified both biases. [4, 5, 6, 7] Column standardization removes target offsets, but it leaves the per-ligand shared direction, which dominates the target-correlation spectrum. Calling this a general binding axis would imply a biological universality that the docking scores do not establish.

Two-way centering changes the scientific object. A column-standardized PR dimension of 1.83 did not leave only two residual directions: the same matrix had residual PR dimension 9.30.

DOCKSTRING changed from 2.27 to 18.21. These increases persisted after normalization by the respective algebraic maxima, throughout target subsampling, and in the 500-permutation parallel analysis. Thus, concentration of the column-standardized surface cannot support a claim that Vina output lacks target-specific variation. It supports a narrower warning: analyses that treat uncentered target columns as independent mechanistic descriptors can be dominated by a shared ligand direction.

The ChEMBL comparison is supporting evidence on matched support, not a general measurement of experimental binding dimensionality. Exact-relation median aggregation, human binding restrictions and a Ki/Kd-only block all retained higher PR dimension than docking, but the panels contained only six aminergic targets, mixed laboratories and conditions, and imputed cells. The larger incomplete block was sensitive to the missing-data estimator. These controls narrow the interpretation to the observed matched ChEMBL surfaces.

Several further limits determine scope. Both large replications use Vina-family docking; the RF-Score v1 panels contain only 35–41 ligands and cannot stand in for a large modern neural scoring matrix. Target-family labels are broad curation categories, and neither target panel is a probability sample from all pharmacologically relevant proteins. Leave-one-out ranges and target-subset distributions therefore measure composition sensitivity within the two observed panels rather than superpopulation uncertainty. The scaffold bootstrap likewise uses one seeded DOCKSTRING support. An exploratory rank–selectivity analysis of 20 dependent, heterogeneous DTI arms is confined to Supplementary Fig. S3 and Tables S5–S7; its dependence on an outcome-related 12-arm restriction is why it is not part of the central claim.

A practical spectral audit should report both the column-standardized and two-way-centered residual spectra, use transformation-matched empirical nulls, and state the target and chemical populations to which uncertainty applies. The PR effective dimension describes variance concentration. It does not determine whether the residual geometry is biophysically correct or predictively useful.

4 Conclusions

The two large tested Vina docking-score matrices are strongly spectrally concentrated after column standardization, with PR effective dimensions 1.83 and 2.27. This result is stable within the observed chemical and target panels, but it does not generalize automatically to other docking engines or learned scoring functions. Removing ligand and target main effects raises the PR dimensions to 9.30 and 18.21, showing that aggregate concentration coexists with substantially richer two-way-centered residual structure.

5 Methods

5.1 Datasets and score panels

The antitarget matrix was taken from the open resource of Nikitin et al.[10] and contained 12,651 analyzable compounds and 44 Vina-GPU target scores.[16] The source pipeline clipped positive scores to zero; its processed matrix contained 111 such zero-censored cells (0.020%) and no remaining positive values. Missing entries comprised 1.466% of cells and were replaced by the target mean for the primary analysis. Chemical-cluster uncertainty used the precomputed 8,150 Butina clusters supplied with the dataset. The three-target expansion added BSEP, FXR and PXR scores generated under the same project protocol.

DOCKSTRING was read from its released full score table.[9] We used the 58 target columns, clipped positive scores to zero for consistency and used the 260,060 rows with complete scores for the primary spectrum. Only 0.0022% of entries were missing and 0.0359% were positive before clipping.

RF-Score v1 was evaluated with the Open Drug Discovery Toolkit implementation.[11, 14] The first panel comprised the 41 ligands complete across RF-Score, smina/Vina and original Vina scores for 14 targets. The second comprised 35 ligands complete across RF-Score on Vinardo poses and original Vina scores for 20 targets. These are scoring-function and pose-source sensitivities, not affinity benchmarks.

Experimental activities came from ChEMBL release 34.[22] We retained records with a pChEMBL value and standard type Ki, Kd, IC50 or EC50; canonical SMILES were converted to full InChIKeys and targets were mapped to the panel's single-protein accessions. The primary matched block was rebuilt from activity-level records, required an exact standard relation and full InChIKey, and aggregated repeated compound–target activities by the median. It used 74 compounds with docking scores for six targets and ChEMBL measurements for at least five targets; 9.9% of experimental cells were median-imputed within target. We additionally restricted to human binding assays, either with all four endpoints or with Ki/Kd only. A secondary 212-by-22 block required at least eight measured targets per ligand and 30 measurements per retained target. We compared five-nearest-neighbor imputation, multiple imputation by chained equations, low-rank matrix completion and probabilistic-PCA expectation maximization. Assay sensitivity separately retained binding records with Ki or Kd endpoints from the already human-only curated refetch. Relations reported as inequalities and records without pChEMBL were excluded. A frozen table that retained the most potent pChEMBL value per compound–target pair was analyzed only as a legacy sensitivity with a paired ligand bootstrap.

DAVIS experimental affinity[13] and Boltz-2[12] were read from versioned matrices in the release only for the contextual centering ladder in Fig. 5. The common surface contained 72 ligands and 442 kinases. PR dimensions were computed on each complete dense matrix, including pKd-floor values, because the surfaces were aligned. The exploratory rank–selectivity analysis is reported entirely in the Supplement; its complete panel has 20 arms, while the 12-arm above-chance subset is an outcome-restricted sensitivity.

5.2 Spectral quantities

For each complete matrix X, columns with zero variance were removed and the remaining columns were standardized to zero mean and unit sample variance, producing Z. We formed the target correlation matrix

$$C = \frac{1}{N-1} Z^{\mathsf{T}} Z \tag{4}$$

and sorted its non-negative eigenvalues $\lambda_1 \geq \cdots \geq \lambda_P$. The primary summary was the PR effective dimension in Eq. 2. We also computed the entropy rank $\exp[-\sum_k p_k \log p_k]$, where $p_k = \lambda_k / \sum_l \lambda_l$, the fraction explained by principal component 1 and the number of components reaching 80% and 90% cumulative variance. Equation 3 follows from $\mathrm{tr}(C^2) = P + P(P-1)\overline{r_{ij}^2}$. The PR effective dimension is especially sensitive to dominant eigenvalues, whereas entropy rank weights the spectral tail more strongly. Both are continuous summaries of variance concentration, not algebraic or nonlinear manifold ranks.

The Marchenko–Pastur upper edge $(1 + \sqrt{P/N})^2$ was used as an asymptotic guide. The original panel analyses anchored mode counts on column-wise permutation parallel analysis, which preserves each score marginal while breaking cross-target alignment.[15] For direct column-standardized–residual comparison, we additionally drew the same 12,000 rows from each large matrix, generated 500 independent column-wise permutations, applied either no centering or two-way centering to both observed and permuted matrices, and counted observed eigenvalues above the rank-wise 95th percentile of the corresponding null. We repeated the count in five independent 100-permutation series to assess Monte Carlo stability. The 44-target analysis additionally used cluster-preserving and cluster-representative nulls; the latter retained three leading modes and controlled the inflated false-positive rate caused by chemical clustering. Effective-dimension intervals for that panel resampled Butina clusters. DOCKSTRING used both molecule resampling and a Bemis–Murcko scaffold-cluster bootstrap on one seeded 15,000-row support. The legacy matched experiment used a paired compound bootstrap. These intervals are conditional on the observed target panels; chemical resampling does not represent target-selection uncertainty.

5.3 Main-effect removal

For complete matrices we computed the interaction residual in one closed-form step,

$$\Gamma = X - \bar{X}_{\cdot j} - \bar{X}_{i\cdot} + \bar{X}_{\cdot\cdot}, \tag{5}$$

where the bars denote column, row and grand means. We then column-standardized Γ and computed exactly the same spectral summaries. Because standardization already removes column offsets, the change from the column-standardized to residual spectra is driven principally by removal of the per-ligand common direction. The fraction of grand-centered unstandardized sums of squares retained by Γ was reported separately and was not used as a PR-dimension measure. Each row of Γ sums to zero. Subsequent non-zero column scaling preserves one exact linear dependence, so $\mathrm{rank}(\Gamma) \leq P - 1$ and the residual correlation spectrum contains one structural zero eigenvalue. For an additive model with independent, identically distributed residual noise, the non-zero residual spectrum is flat and the PR dimension approaches $P - 1$. We therefore treat $P - 1$ as a noise ceiling, not as a panel-size correction, and show target-subsampling curves for the residual estimand.

5.4 Physicochemical-axis analyses

The full-panel and matched-support residualizations used seven RDKit descriptors: heavy-atom count, molecular weight, Labute accessible surface area, topological polar surface area, Crippen cLogP, rotatable-bond count and ring count. Descriptors were standardized. For the full anti-target panel, a separate ordinary least-squares model was fit for each target and residuals were assembled before spectral analysis. The nonlinear sensitivity used five-fold shuffled cross-fitting (seed 0) with one histogram gradient-boosting regressor per target (200 iterations, learning rate 0.05, maximum depth 4); no ligand's fitted value came from a model trained on that ligand. The same protocol was repeated on the matched 74-by-6 support. The narrow-size sensitivity retained molecules with 17–22 heavy atoms, and ligand efficiency divided each score by heavy-atom count. A separate cached diagnostic of the shared score direction used molecular weight, cLogP, topological polar surface area, hydrogen-bond donor and acceptor counts, rotatable-bond count and molecular volume. Pocket volumes were read from the frozen 44-pocket structure summary; 36 targets had complete volumes. The reported univariate statistic correlates pocket volume with the absolute correlation between each target score and the per-ligand panel mean, not with a principal-component loading. The 13-descriptor pocket model was evaluated by leave-one-target-out prediction.

5.5 Preprocessing and experimental-noise sensitivities

For the antitarget panel we repeated the raw analysis with target-median imputation ($r_{\mathrm{PR}} = 1.835$), complete cases (9,297 ligands; 1.320), pairwise-complete Pearson correlations (1.788) and pairwise Spearman correlations (1.686). The complete-case estimate changes the chemical support and is not a like-for-like imputation test. Censoring sensitivities treated source-clipped zero cells as missing, removed every affected ligand or removed the most affected target decile. For interaction spectra we compared raw-scale centering followed by standardization (9.303), standardization followed by centering (9.378), and robust median/interquartile-range scaling followed by centering (9.462). These analyses test analysis-pipeline sensitivity rather than define alternative primary endpoints.

Different-document ChEMBL repeats for the same compound–target pair provided a conservative repeatability scale. From 279,012 paired records, the median absolute difference was 0.47 pChEMBL and the implied single-measurement Gaussian standard deviation was 0.493. For each of the six matched targets, this standard deviation was divided by that target's experimental standard deviation. Independent Gaussian noise at the resulting target-specific scales was added to the standardized docking block in 2,000 simulations before recomputing PR dimension. This calibration includes inter-laboratory and condition variation; it is not a pure technical replicate variance or a complete experimental noise model.

5.6 Chemical- and target-composition sensitivities

Target-subsampling curves used 200 subsets without replacement at each reported size on one seeded 12,000-ligand support; the complete-panel points used all eligible ligands. Uniform sampling was paired with broad family-stratified sampling based on the versioned `data/target_families.csv` manifest. Stratified allocations were proportional to family size and included at least one target from every family whenever the requested subset size permitted. Both the column-standardized and two-way-centered residual PR dimensions were recomputed after every draw.

Leave-one-target-out analysis used all 12,651 Docking-44 ligands and one seeded 15,000-row DOCKSTRING support; complete-panel estimates on the same support were plotted as references. The resulting minima, medians and maxima are composition sensitivities, not confidence intervals for a target superpopulation. Ligand-subsampling curves used 50 subsets. For the DOCKSTRING chemical-cluster sensitivity, Bemis–Murcko scaffolds were generated with RDKit; acyclic molecules were treated as singletons, and 200 cluster-bootstrap replicates were compared with 200 molecule-bootstrap replicates on the same 15,000-molecule support. All random procedures used seed 0.

5.7 Software and reproducibility

The new analyses and figures were run with Python 3.12.11, NumPy 1.26.4, pandas 2.3.2, SciPy 1.16.2, scikit-learn 1.7.2, Matplotlib 3.10.6 and RDKit 2025.03.6. The release's `analysis/build_evidence.py` regenerates the centering ladder, preprocessing controls, experimental sensitivities, subsampling results and DTI associations in `results/evidence_summary.json`; it also embeds explicitly labelled summaries from the frozen upstream docking/rescoring stages. `analysis/make_figures.py` regenerates all main and supplementary figures. The core release reproduces the reported analysis from frozen matrices and summaries; it does not rerun receptor preparation, docking or RF-Score pose rescoring. A checksum manifest, pinned requirements, environment file, Dockerfile and Makefile provide input verification and one-command regeneration. Random seed 0 is fixed throughout.

OpenAI Codex and Anthropic Claude were used during manuscript preparation for code review, literature-search assistance and language editing. They were not treated as sources of scientific evidence. The authors inspected the underlying data and code, verified the reported claims and citations, and retain responsibility for the work.

List of abbreviations

DTI, drug–target interaction; KNN, k-nearest neighbors; PC, principal component; RMT, random matrix theory; PR, participation ratio.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The source antitarget dataset is available with Nikitin et al.[10]; DOCKSTRING scores and poses are available from the DOCKSTRING release.[9] DAVIS and ChEMBL are publicly available from their respective source repositories. The derived matrices, frozen summaries, source manifest and analysis code are available without registration under an OSI-approved code license at `https://github.com/AdeliyaLeleytner/vina-score-spectral-concentration`. The exact `v1.0.0` release will be archived at **[ZENODO DOI]**. Upstream datasets retain their stated licenses and are identified by source URL, version and checksum in the manifest. The DOI placeholder must be replaced by the archival identifier before submission.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

A.L. conceived the study, performed the analyses and drafted the manuscript. V.S. and M.F. contributed to study design and interpretation. All authors reviewed the manuscript and are responsible for the final submitted version.

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